

DPP - 7

- Q.1 Suppose families always have one, two or three children, with probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$ respectively. Assume everyone eventually gets married and has children, the probability of a couple having exactly four grandchildren is
- (A) $\frac{27}{128}$ (B) $\frac{37}{128}$ (C) $\frac{25}{128}$ (D) $\frac{20}{128}$
- Q.2 Miss C has either Tea or Coffee at morning break. If she has tea one morning, the probability she has tea the next morning is 0.4. If she has coffee one morning, the probability she has coffee next morning is 0.3. Suppose she has coffee on a Monday morning. The probability that she has tea on the following Wednesday morning is
- (A) 0.46 (B) 0.49 (C) 0.51 (D) 0.61
- Q.3 In a maths paper there are 3 sections A, B & C. Section A is compulsory. Out of sections B & C a student has to attempt any one. Passing in the paper means passing in A & passing in B or C. The probability of the student passing in A, B & C are p, q & $\frac{1}{2}$ respectively. If the probability that the student is successful is $\frac{1}{2}$ then :
- (A) $p = q = 1$ (B) $p = q = \frac{1}{2}$ (C) $p = 1, q = 0$ (D) $p = 1, q = \frac{1}{2}$
- Q.4 A box contains 100 tickets numbered 1, 2, 3, ..., 100. Two tickets are chosen at random. It is given that the maximum number on the two chosen tickets is not more than 10. The minimum number on them is 5, with probability
- (A) $\frac{1}{9}$ (B) $\frac{2}{11}$ (C) $\frac{3}{19}$ (D) none
- Q.5 Sixteen players $s_1, s_2, \dots, s_{16}$ play in a tournament. They are divided into eight pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength. The probability that "exactly one of the two players s_1 & s_2 is among the eight winners" is
- (A) $\frac{4}{15}$ (B) $\frac{7}{15}$ (C) $\frac{8}{15}$ (D) $\frac{9}{15}$
- Q.6 The number 'a' is randomly selected from the set $\{0, 1, 2, 3, \dots, 98, 99\}$. The number 'b' is selected from the same set. Probability that the number $3^a + 7^b$ has a digit equal to 8 at the units place, is
- (A) $\frac{1}{16}$ (B) $\frac{2}{16}$ (C) $\frac{4}{16}$ (D) $\frac{3}{16}$
- Q.7 On a normal standard die one of the 21 dots from any one of the six faces is removed at random with each dot equally likely to be chosen. The die is then rolled. The probability that the top face has an odd number of dots is
- (A) $\frac{5}{11}$ (B) $\frac{5}{12}$ (C) $\frac{11}{21}$ (D) $\frac{6}{11}$

- Q.8 Two boys A and B find the jumble of n ropes lying on the floor. Each takes hold of one loose end randomly. If the probability that they are both holding the same rope is $\frac{1}{101}$ then the number of ropes is equal to
(A) 101 (B) 100 (C) 51 (D) 50

[REASONING TYPE]

- Q.9 A fair coin is tossed 3 times consider the events
A : first toss is head
B : second toss is head
C : exactly two consecutive heads or exactly two consecutive tails.
Statement-1: A, B, C are independent events.
because
Statement-2: A, B, C are pairwise independent.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- Q.10 Let a sample space S contains n elements. Two events A and B are defined on S, and $B \neq \phi$.
Statement-1: The conditional probability of the event A given B, is the ratio of the number of elements in AB divided by the number of elements in B.
because
Statement-2: The conditional probability model given B, is equally likely model on B.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- Q.11 A bag contains 6 balls of 3 different colours namely White, Green and Red, atleast one ball of each different colour. Assume all possible probability distributions are equally likely.
(a) The probability that the bag contains 2 balls of each colour, is
(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) $\frac{1}{10}$ (D) $\frac{1}{4}$
(b) Three balls are picked up at random from the bag and found to be one of each different colour. The probability that the bag contained 4 Red balls is
(A) $\frac{1}{14}$ (B) $\frac{2}{14}$ (C) $\frac{3}{14}$ (D) $\frac{4}{14}$
(c) Three balls are picked at random from the bag and found to be one of each different colour. The probability that the bag contained equal number of White and Green balls, is
(A) $\frac{4}{14}$ (B) $\frac{3}{14}$ (C) $\frac{2}{14}$ (D) $\frac{5}{14}$

- Q.12 Two fair dice are rolled. Let $P(A_i) > 0$ denotes the event that the sum of the number appearing on the faces of the dice is divisible by i .
- (a) Which one of the following events is most probable?
 (A) A_3 (B) A_4 (C) A_5 (D) A_6
- (b) For which one of the following pairs (i, j) are the events A_i and A_j are independent?
 (A) $(3, 4)$ (B) $(4, 6)$ (C) $(2, 3)$ (D) $(4, 2)$
- (c) Number of all possible ordered pairs (i, j) for which the events A_i and A_j are independent.
 (A) 6 (B) 12 (C) 13 (D) 25
- Q.13 A multiple choice test question has five alternative answers, of which only one is correct. If a student has done his home work, then he is sure to identify the correct answer; otherwise, he chooses an answer at random. Let E : denotes the event that a student does his home work with $P(E) = p$ and F : denotes the event that he answer the question correctly.
- (a) If $p = 0.75$ the value of $P(E/F)$ equals
 (A) $\frac{8}{16}$ (B) $\frac{10}{16}$ (C) $\frac{12}{16}$ (D) $\frac{15}{16}$
- (b) The relation $P(E/F) \geq P(E)$ holds good for
 (A) all values of p in $[0, 1]$ (B) all values of p in $(0, 1)$ only
 (C) all values of p in $[0.5, 1]$ only (D) no value of p .
- (c) Suppose that each question has n alternative answers of which only one is correct, and p is fixed but not equal to 0 or 1 then $P(E/F)$
 (A) decreases as n increases for all $p \in (0, 1)$
 (B) increases as n increases for all $p \in (0, 1)$
 (C) remains constant for all $p \in (0, 1)$
 (D) decreases if $p \in (0, 0.5)$ and increases if $p \in (0.5, 1)$ as n increases

[MULTIPLE OBJECTIVE TYPE]

- Q.14 A boy has a collection of blue and green marbles. The number of blue marbles belong to the sets $\{2, 3, 4, \dots, 13\}$. If two marbles are chosen simultaneously and at random from his collection, then the probability that they have different colour is $1/2$. Possible number of blue marbles is :
 (A) 2 (B) 3 (C) 6 (D) 10
- Q.15 If A & B are two events such that $P(B) \neq 1$, B^c denotes the event complementary to B , then
 (A) $P(A/B^c) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$
 (B) $P(A \cap B) \geq P(A) + P(B) - 1$
 (C) $P(A) > < P(A/B)$ according as $P(A/B^c) > < P(A)$
 (D) $P(A/B^c) + P(A^c/B^c) = 1$
- Q.16 For $P(A) = \frac{3}{8}$; $P(B) = \frac{1}{2}$; $P(A \cup B) = \frac{5}{8}$ which of the following do/does hold good?
 (A) $P(A^c/B) = 2P(A/B^c)$ (B) $P(B) = P(A/B)$
 (C) $15P(A^c/B^c) = 8P(B/A^c)$ (D) $P(A/B^c) = P(A \cap B)$

- Q.17 If E_1 and E_2 are two events such that $P(E_1) = 1/4$, $P(E_2/E_1) = 1/2$ and $P(E_1/E_2) = 1/4$
 (A) then E_1 and E_2 are independent
 (B) E_1 and E_2 are exhaustive
 (C) E_2 is twice as likely to occur as E_1
 (D) Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in G.P.
- Q.18 Two events A and B are such that the probability that at least one of them occurs is $5/6$ and both of them occurring simultaneously is $1/3$. If the probability of not occurrence of B is $1/2$ then
 (A) A and B are equally likely (B) A and B are independent
 (C) $P(A/B) = 2/3$ (D) $3 P(A) = 4 P(B)$
- Q.19 The probabilities of events, $A \cap B$, A, B & $A \cup B$ are respectively in A.P. with probability of second term equal to the common difference. Therefore the events A and B are
 (A) mutually exclusive (B) independent
 (C) such that one of them must occur (D) such that one is twice as likely as the other
- Q.20 A box contains 11 tickets numbered from 1 to 11. Six tickets are drawn simultaneously at random. Let E_1 denotes the event that the sum of the numbers on the tickets drawn is even and E_2 denotes the event that the sum of the numbers on the tickets drawn is odd. Which of the following hold good?
 (A) E_1 and E_2 are equally likely (B) E_1 and E_2 are exhaustive
 (C) $P(E_2) > P(E_1)$ (D) $P(E_1/E_2) = P(E_2/E_1)$
- Q.21 If $\bar{E}$ & $\bar{F}$ are the complementary events of events E & F respectively & if $0 < P(F) < 1$, then :
 (A) $P(E|F) + P(\bar{E}|F) = 1$ (B) $P(E|F) + P(E|\bar{F}) = 1$
 (C) $P(\bar{E}|F) + P(E|\bar{F}) = 1$ (D) $P(E|\bar{F}) + P(\bar{E}|\bar{F}) = 1$
- Q.22 Probability of n heads in $2n$ tosses of a fair coin can be given by
 (A) $\prod_{r=1}^n \left(\frac{2r-1}{2r} \right)$ (B) $\prod_{r=1}^n \left(\frac{n+r}{2r} \right)$ (C) $\sum_{r=0}^n \left(\frac{{}^nC_r}{2^n} \right)^2$ (D) $\frac{\sum_{r=0}^n ({}^nC_r)^2}{\left(\sum_{r=0}^n {}^nC_r \right)^2}$
- Q.23 Which of the following statements is/are True?
 (A) A fair coin is tossed n times where n is a positive integer. The probability that n^{th} toss results in head is $1/2$.
 (B) The conditional probability that the n^{th} toss results in head given that first $(n-1)$ tosses results in head is $1/2^n$
 (C) Let E and F be the events such that F is neither impossible nor sure.
 If $P(E/F) > P(E)$ then $P(E/F^c) > P(E)$
 (D) If A, B and C are independent then the events $(A \cup B)$ and C are independent.

[MATCH THE COLUMN]

Q.24

Column-I

Column-II

- | | | |
|-----|---|--------|
| (A) | Two different numbers are taken from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The probability that their sum and positive difference, are both multiple of 4, is $x/55$ then x equals | (P) 4 |
| (B) | There are two red, two blue, two white and certain number (greater than 0) of green socks in a drawer. If two socks are taken at random from the drawer without replacement, the probability that they are of the same colour is $1/5$ then the number of green socks are | (Q) 6 |
| (C) | A drawer contains a mixture of red socks and blue socks, at most 17 in all. It so happens that when two socks are selected randomly without replacement, there is a probability of exactly $1/2$ that both are red or both are blue. The largest possible number of red socks in the drawer that is consistent with this data, is | (R) 8 |
| | | (S) 10 |

Q.25

Column-I

Column-II

- | | | |
|-----|--|-------|
| (A) | In a knockout tournament 2^n equally skilled players; $S_1, S_2, \dots, S_{2^n}$ are participating. In each round players are divided in pair at random and winner from each pair moves in the next round. If S_2 reaches the semifinal then the probability that S_1 wins the tournament is $\frac{1}{20}$. The value of 'n' equals | (P) 3 |
| (B) | In a multiple choice question there are four alternative answers of which one or more than one is correct. A candidate will get marks on the question only if he ticks all the correct answers. The candidate ticks the answers at random. If the probability of the candidate getting marks on the question is to be greater than or equal to $1/3$ the least number of chances he should be allowed is | (Q) 4 |
| (C) | All the face cards from a pack of 52 playing cards are removed. From the remaining pack half of the cards are randomly removed without looking at them and then randomly drawn two cards simultaneously from the remaining. If the probability that, two cards drawn are both aces, is $\frac{p({}^{38}C_{20})}{{}^{40}C_{20} \cdot {}^{20}C_2}$, then the value of p is | (R) 5 |
| (D) | An unbiased normal coin is tossed 'n' times. Let
E_1 : event that both Heads and Tails are present in 'n' tosses.
E_2 : event that the coin shows up Heads atmost once.
The value of 'n' for which E_1 and E_2 are independent, is | (S) 6 |